25900514 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMATA01 - INTRODUCTION TO DATA ANALYSIS USING PYTHON

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer any **10** questions

Each question carries **3** marks

1. Which keyword is used for handling exceptions in Python? [A] / [CO1]

   a). try b). error c). while d). for

2. What is the output of the following code? [A] / [CO2]

```
import numpy as np
a = np.array([[1,2],[3,4]])
b = np.array([[5,6],[7,8]])
print(a + b)
```

   a). [[6,8],[10,12]] b). [[5,6],[7,8]] c). [[1,2],[3,4]] d). Error

3. Library for skewness/kurtosis: [U] / [CO3]

   a). math b). stat c). scipy.stats d). random

4. Consider the observed frequencies [12, 8] from tossing a coin 20 times. The expected counts are [10, 10]. What is the chi-square value (rounded to 2 decimals)? [E] / [CO4]

   a). 0.4 b). 0.8 c). 0.6 d). 0.5

5. Which among the following function is used to remove first occurrence of a value in an array? [A] / [CO2]

a). delete() b). remove() c). pop() d). append()

6. What is the output of the following code? [A] / [CO1]

```
x = 25
if  x > 50:
    print("A")
elif x > 20:
    print("B")
else:
    print("C")
```

a). A b). B c). C d). Error

7. Which of the following imports the `math` module in Python? [U] / [CO2]

a). import maths b). import Maths c). import math d). import Math

8. Which of the following accesses the second element of a list named `numbers`? [A] / [CO1]

a). numbers[1] b). numbers[2] c). numbers(1) d). numbers(2)

9. Default index when not provided is: [U] / [CO4]

a). Starts at 0 b). Starts at 1 c). Starts at 2 d). None of the above

10. If `a=10, b=20, c=20` , then the result of `print(c>b and a>b)` is [A] / [CO1]

a). TRUE b). FALSE c). Undefined Error d). None of these

11. Which of the following is **immutable**? [A] / [CO1]

a). List b). Tuple c). Dictionary d). Set

12. Which computes column-wise standard deviation? [An] / [CO4]

a). df.std(axis=1) b). df.std(axis=2) c). df.std() d). df.sd(axis=1)

13. How many points in the following output? [A] / [CO3]

```
x=[1,2,3,4];
y=[2,1,3,0]
plt.scatter(x,y)
```

a). 2 b). 3 c). 4 d). no points

14. What is the code to drop rows with any NaN values: [An] / [CO4]

a). df.drop() b). df.dropna() c). df.fillna() d).df.drop(null=True)

15. Pairplot shows: [U] / [CO3]

a). Pairwise scatter/hist across variables b). Only bar charts c). Only line plots d). Heatmap

[3x10 = 30]

Part B

One or two Sentences

Answer any **4** questions

Each question carries **5** marks

16. Define definite and indefinite loops. Give one Python example for each. [A] / [CO1]

17. Print the current local time in `HH:MM:SS`. [A] / [CO2]

18. Write the code to compute sample variance of [2,4,4,4,5,5,7,9]. [A] / [CO3]

19. Write the code to compute determinant and inverse of a 2x2 matrix. [An] / [CO4]

20. Write the code to print all elements of a 2D list row-wise. [A] / [CO2]

21. Using the built-in `tips` dataset, plot a boxplot of `tip` amounts. [E] / [CO3]

[5x4 = 20]

Part C

Short Essays

Answer any **2** questions

Each question carries **10** marks

22. Explain operator precedence in Python with suitable examples. Why is it important? [A] / [CO1]

23. Write a code to [A] / [CO2]
(a.) Create an array of integers and find its maximum and minimum
(b.) Also find the avaerge of the members in the array

24. Write a python code for Grade distribution pie chart. [An] / [CO3]
Data: A=25, B=30, C=20, D=15, F=10.

25. Create a dataset of 5 students with 'Name' and 'Gender'. Convert 'Gender' into one-hot encoded columns. [E] / [CO4]

[10x2 = 20]